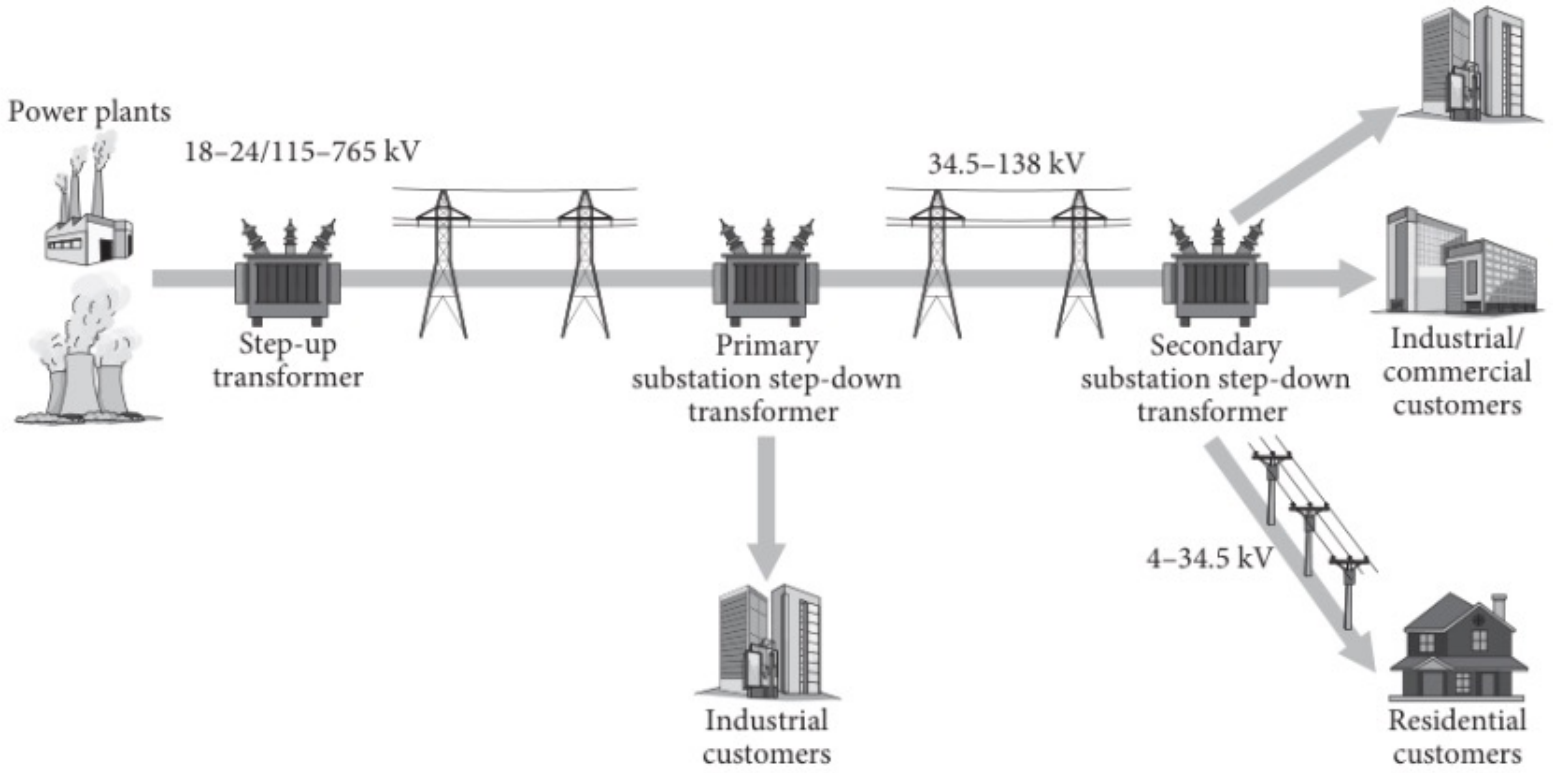
AENL 226
Power Electronics and Power Systems

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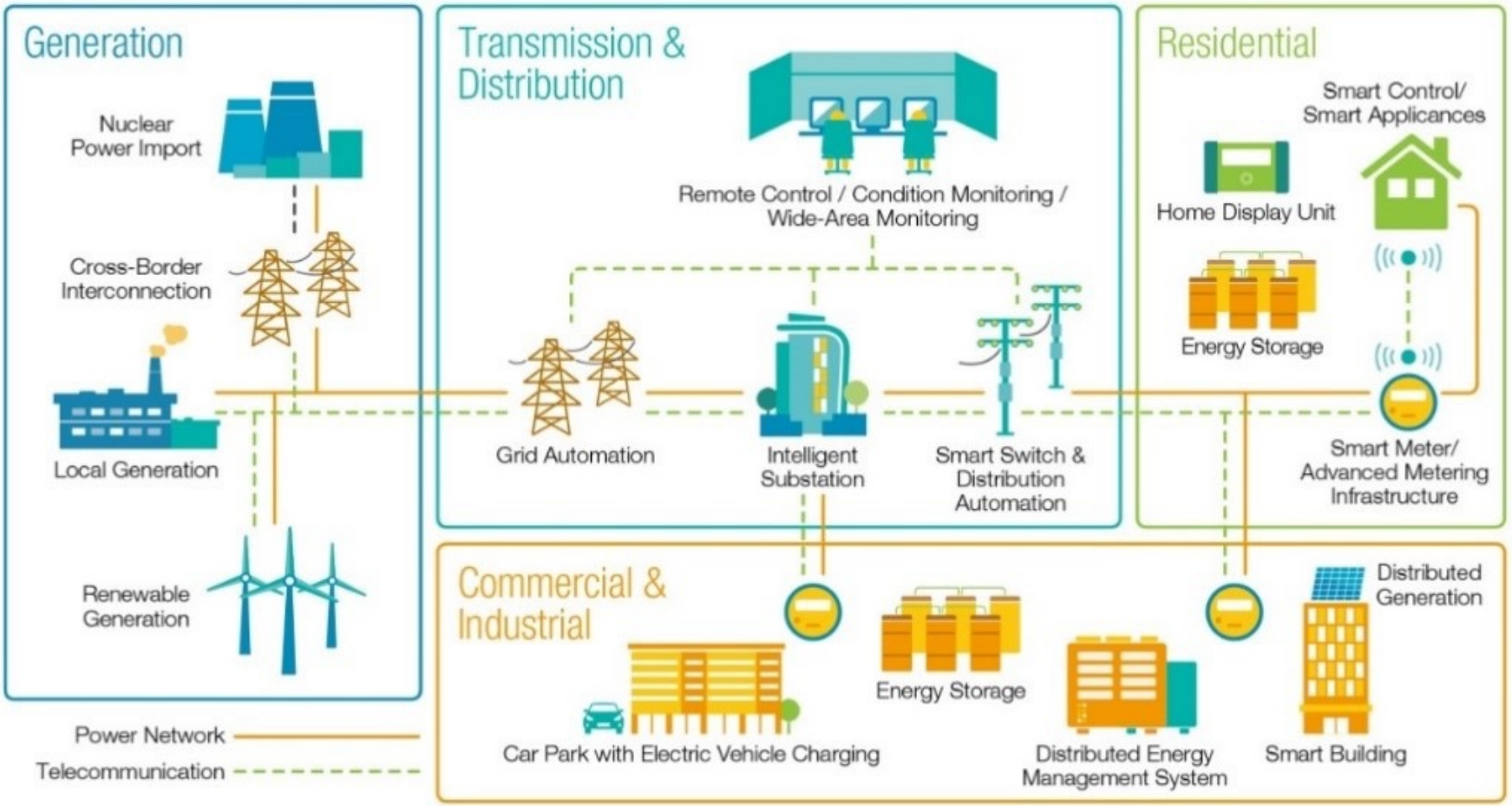


Power System Modeling and Analysis



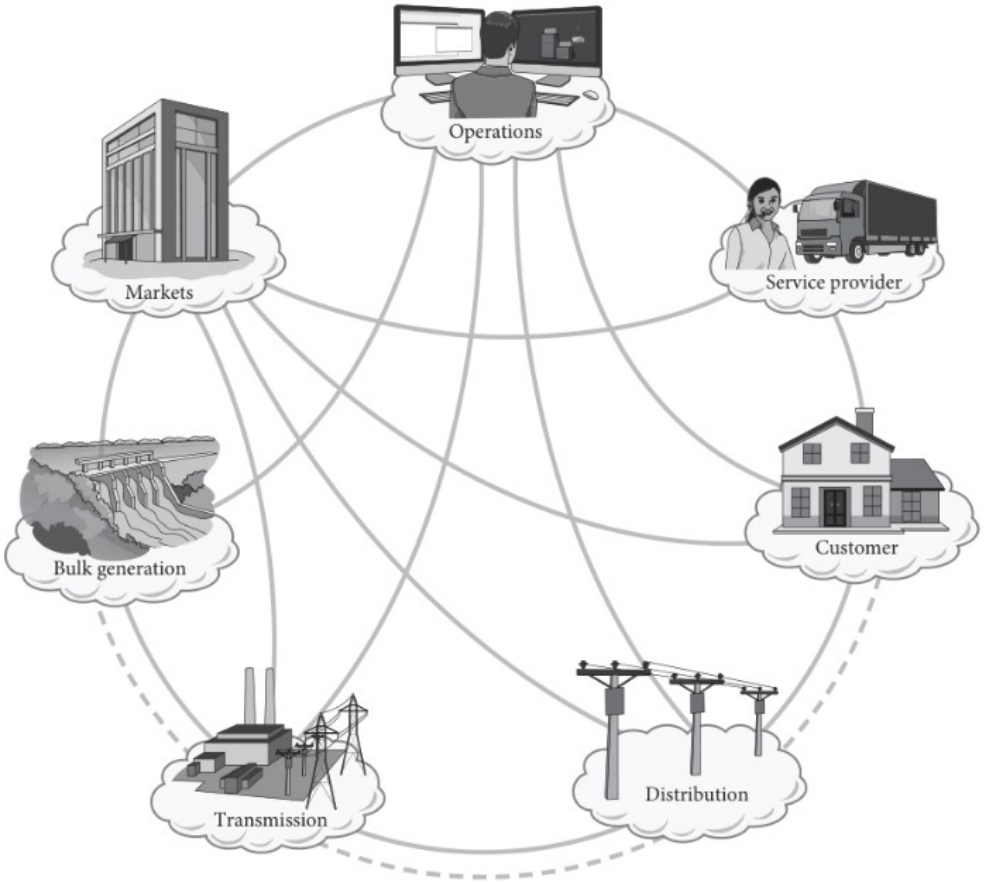
Grainger J. J. G. and Stevenson W. D., Elements of Power System Analysis, McGraw Hill Education, 2017.

Power System Modeling and Analysis



https://www.clp.com.hk/en/about-clp/power-transmission-distribution/smart-grid

Power System Modeling and Analysis



Grainger J. J. G. and Stevenson W. D., Elements of Power System Analysis, McGraw Hill Education, 2017.

Generation

Transmission

Distribution

$$\sum_{i=1}^{NG} P_{Gi} = \sum_{j=1}^{n} P_{Lj} + P_{loss\ total}$$

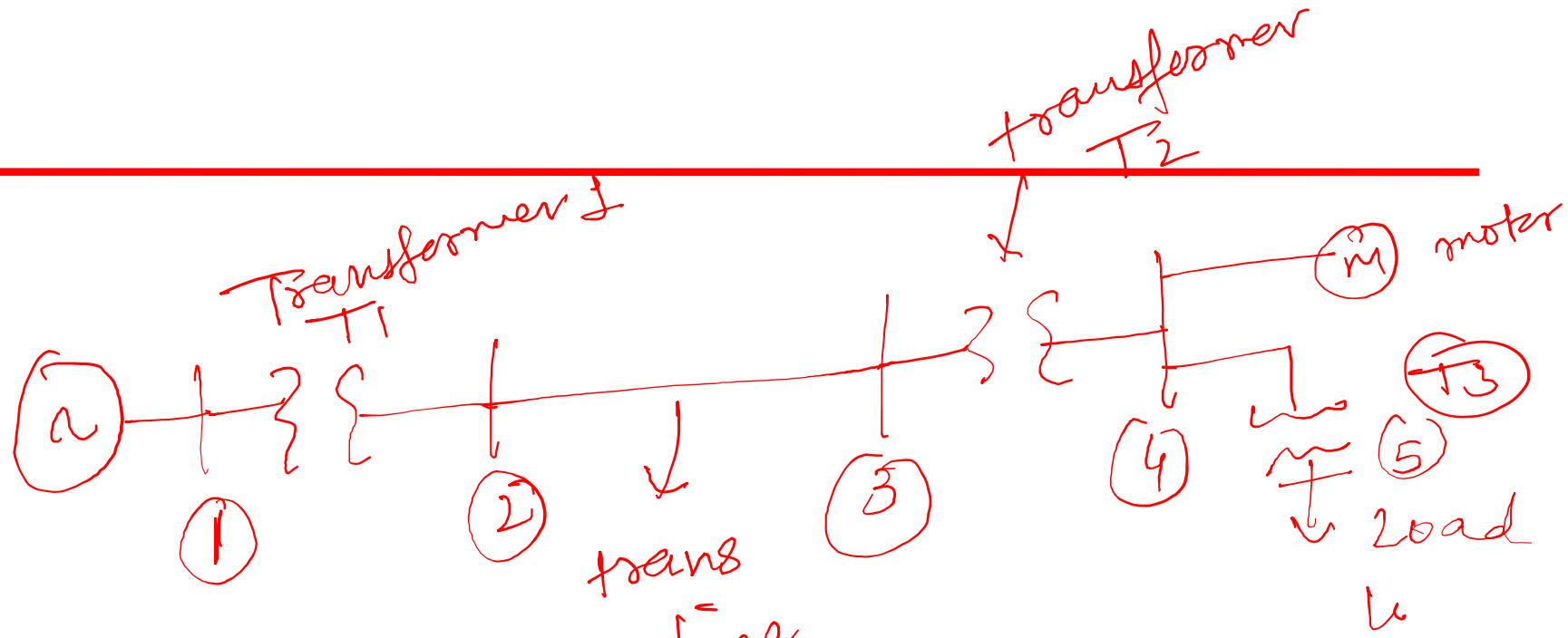
$i = 1, 2 \dots NG$

$j = 1, 2 \dots n$

NG : total number of generators

n : no. of buses in the network (load buses)

any deviation will be reflected in freq.



→ Difference between power transformer & distribution Transformer

→ Single line diagram as given above

Every device

capabilities in terms of active / reactive power, support

loads ← In terms of active / reactive power consumption

Components of Power System ⇒ ?

Generators ⟷ synchronous, Induction, DFIGs

Transformers

Transmission lines

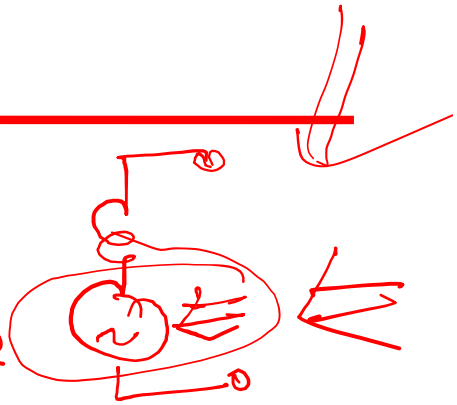
Loads :- motors, heating, electronic loads etc.

⇒ Distribution feeders

Voltage / Power controllers ⇒ STATCOMs, TCSC, SVC

Synchronous Generator ⇒

↳ Controls are possible



there

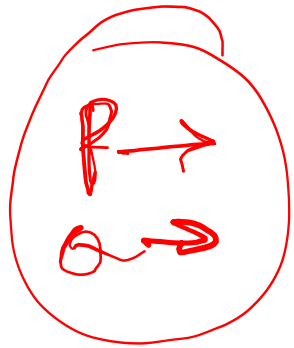
→ real power ⇒ P_{Gi}

voltage ⇒ V_{Gi}

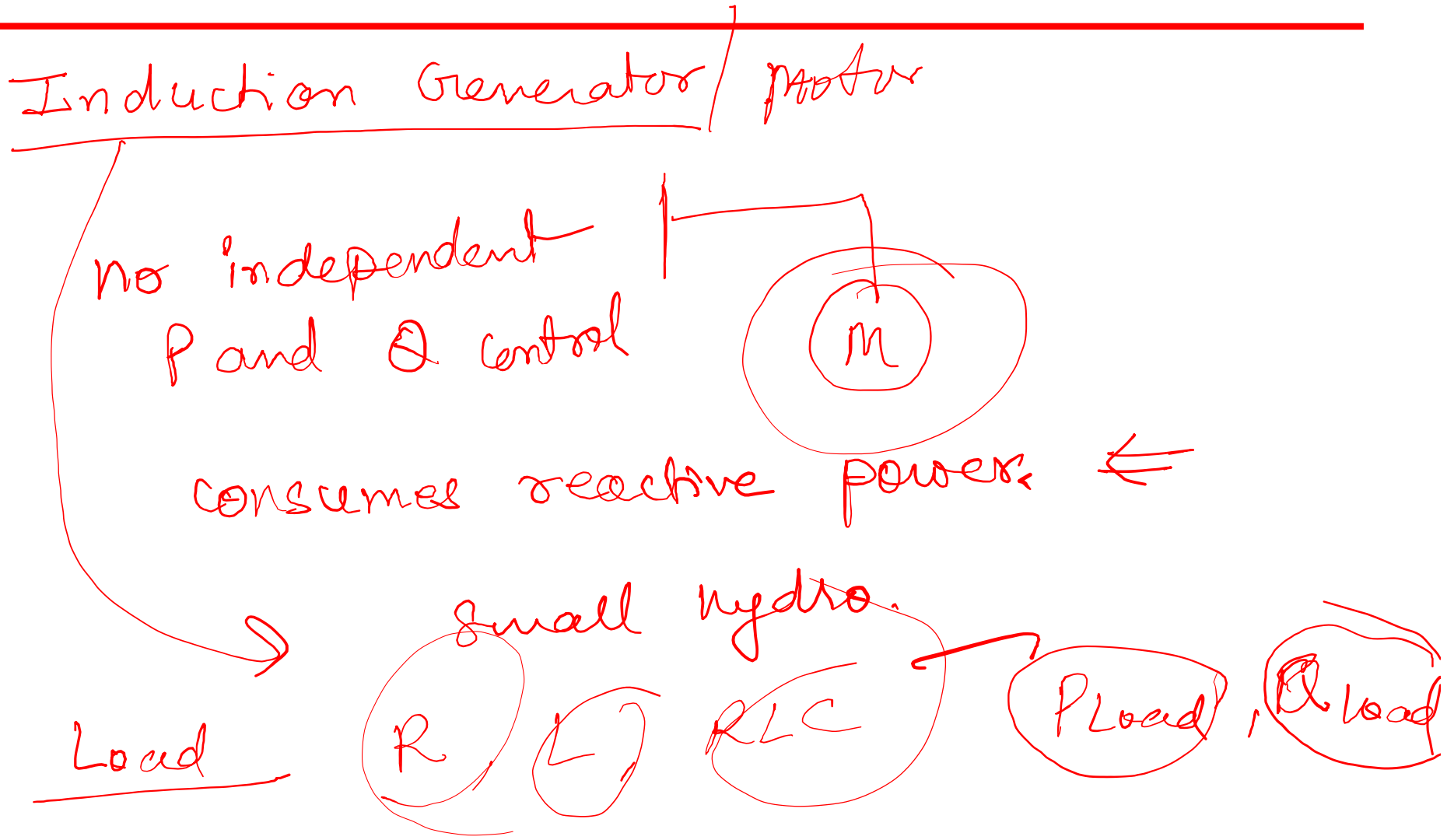
100 mw
of 100 mw
Technical means
30 %

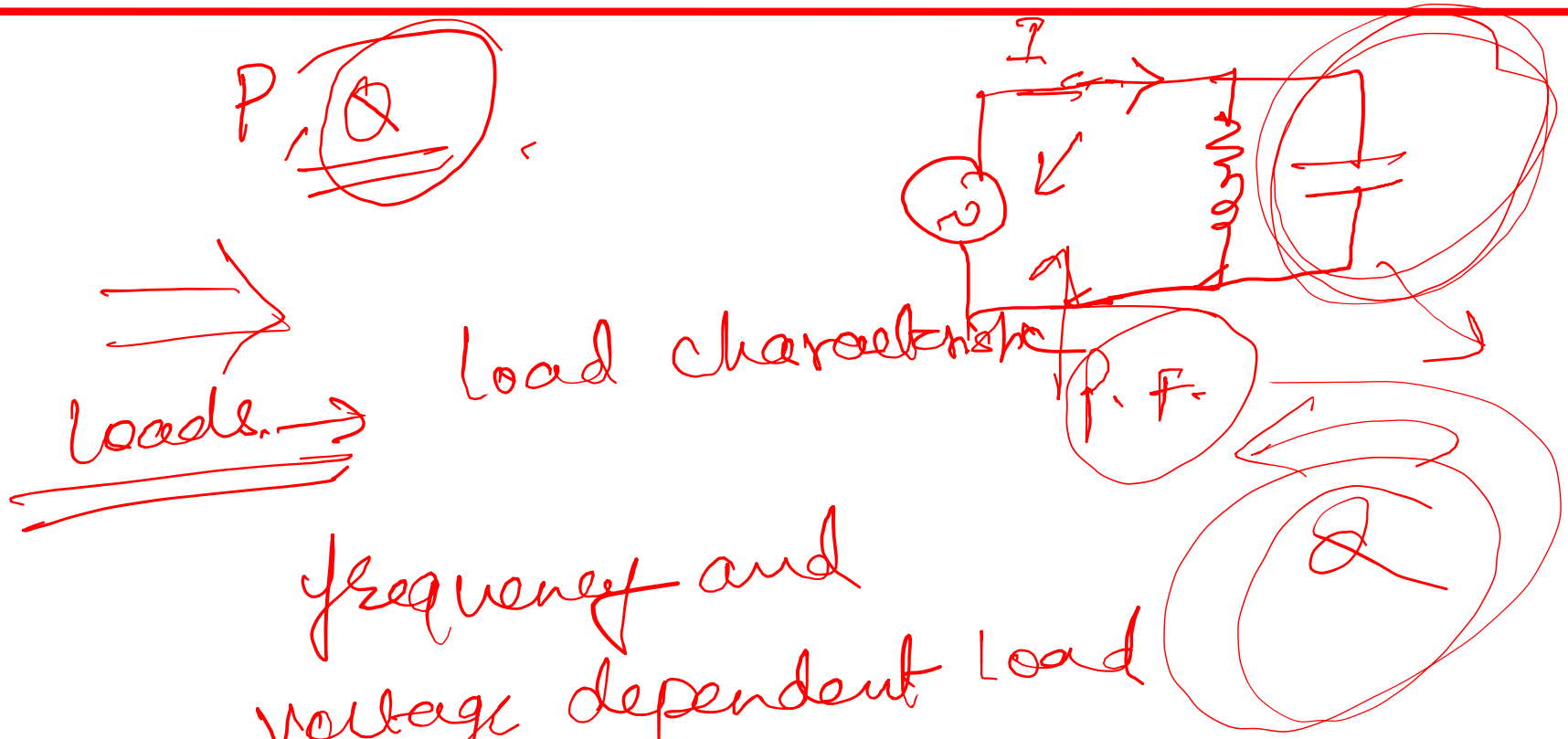
P_{Gi} ⇒ input ⇐

If ⇒ Q_{Gi} ⇒ V_{Gi} = ?



Induction Generator / Motor





Loads → Load characteristic

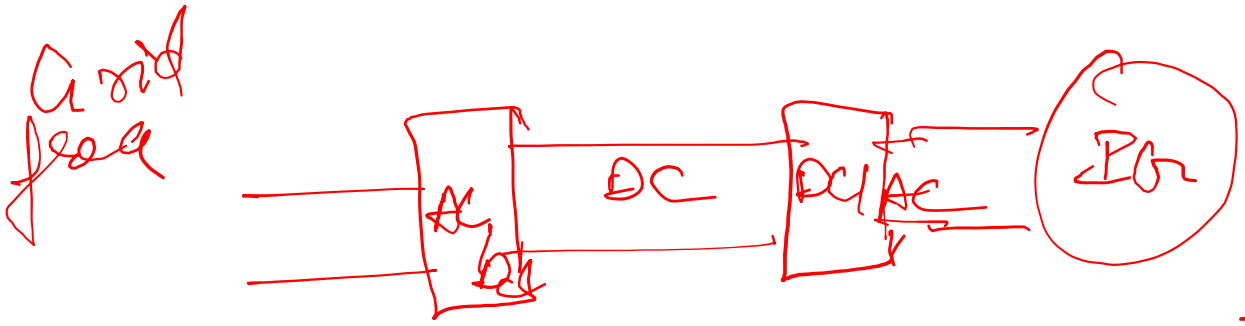
Frequency and voltage dependent Load

Other type of Generators

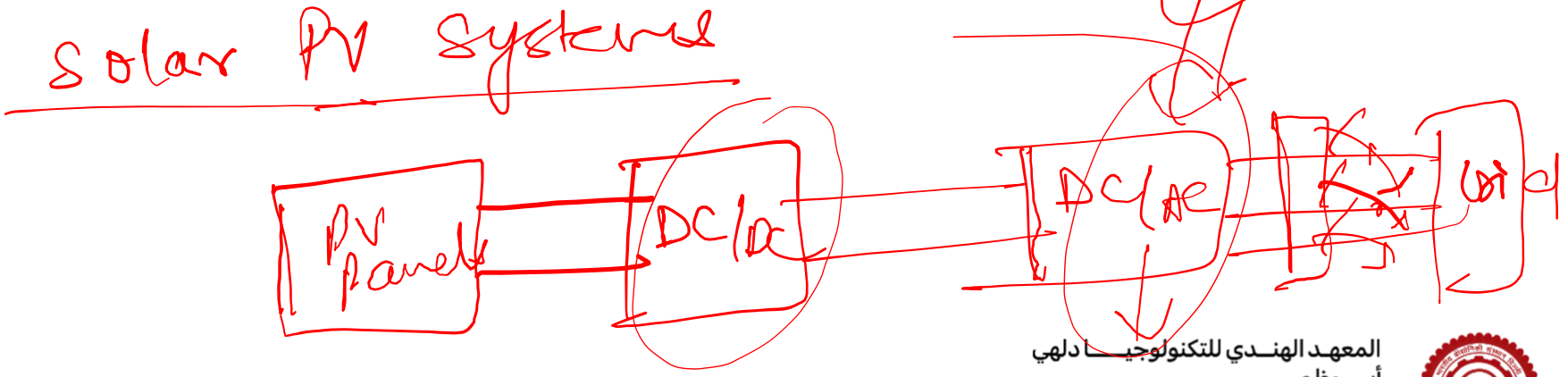
→ Ind^n., PMSG, Wind Energy Conversion System

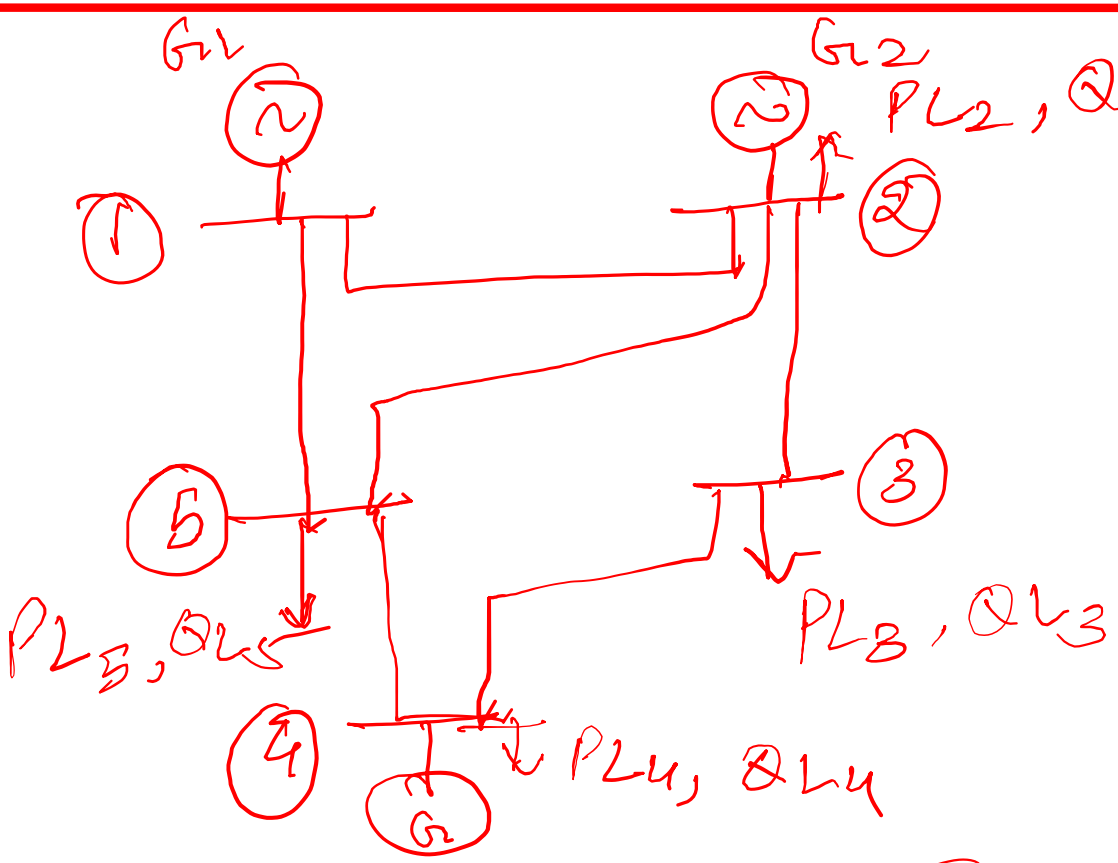
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Doubly fed Ind^{n}. Generators (DFIG)



Solar PV systems





1, 2, 4 ⇒ Gen Bus/PV

3, 5, Load Bus

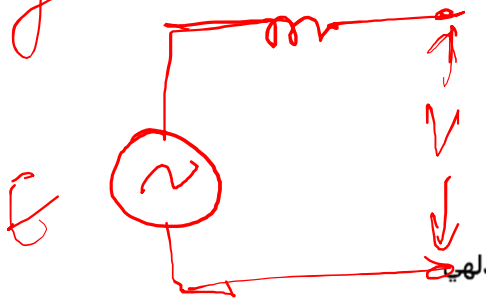
Gen 4 ← Gen-No (3)

⇒ 5-Bus Power System

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$$P_{G_1} + P_{G_2} + P_{G_4} = P_{L_2} + P_{L_3} + P_{L_4} + P_{L_5}$$
$$+ P_{loss_{1-2}} + P_{loss_{2-3}} + P_{loss_{3-4}}$$
$$+ P_{loss_{4-5}} + P_{loss_{5-1}} + P_{loss_{2-5}}$$

Representation of the various components such as generator (asynchronous)



voltage source in series with Xd''

Generators used in thermal, hydro (large), nuclear ⇒ Synchronous gen.

// Independent P & Q control

Generators used in small hydro ⇒ Induction

Wind Energy Conversion Systems

→ Induction gen.

→ Permanent magnet Syn. Gen.

→ DFIG ⇒ Doubly fed Ind. Gen. ✓